



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

A Quick Look to the JavaScript Operators



Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts. Describe applications for an operating system for personal computers in a range of work contexts.

Ask your partner:

- How do you think operators in JavaScript are used?
- What types of operators do you know and what are they used for?

Can Do Statements:

- I can say what an operator is in JavaScript.
- I can list different types of operators in JavaScript.
- I can talk about characteristics of different types of operators in JavaScript.



Let's Explore!

Listen to the following explanation regarding the different operators used in JavaScript.



Instructions: Read the text below.

Before we go deeper into the operators topic, there are some things that have to be cleared out beforehand. In a formula, there are two actors: operands which are the differential factors, and the operators which define the interaction between operands. There are different types of operators people can pick out from to perform a given action in JavaScript. Such operators can be used to perform different operations as in mathematics. Let's check them out.

The arithmetic operators offer a variety of operations which include 'addition', 'subtraction', 'multiplication', 'division' and 'modules', among others. Those operators use a direct arithmetic interaction between operands. The comparison operators offer 'equal', 'not equal', 'greater than' and 'less than' operations and their possible combinations. These operators can be used to check interaction and differences in values between operands. The logical operators like 'logical AND', 'logical OR', and 'logical NOT', have to be used as conditional sentences where a given result depends on a specific condition. The bitwise operators which are 'bitwise AND', 'bitwise OR', 'bitwise XOR', 'left shift', 'right shift', etc., have to be used to perform bit-by-bit operations where values are manipulated to calculate and compare by means of binary numerals. Assignment operators that comprise the 'simple assignment', 'add and assignment', 'subtract and assignment', and others, assign a value to the left operand taking into account the value of the right operand. And the last two which are, on the one hand, the conditional operator that evaluates whether a statement is true or false, and then executes one of these two statements depending on the result; and on the other hand, the "typeof" operator that indicates the data type of the operand.

It is always a good idea to count on a reference chart if you're new at JavaScript. In case you can't get the expected output, you can rely on online tutorials or an experienced person to get you out of trouble. Make sure your spelling is correct to avoid typos and misplaced symbols. This will save you a lot of time instead of having to work something over.





Let's Practice!

Talking About Some Types of Operators in JavaScript

Instructions: Match the sentences from Column A with those in Column B.

COLUMN A

1. The arithmetic operators _____
2. A comparison operator _____
3. The logical operators _____
4. A bitwise operator _____
5. The assignment operators _____

COLUMN B

- a. has to be used to determine if two elements are equal, not equal or greater or smaller than the other.
- b. have to be implemented in order to assign a value to the left operand taking into account the value of the right operand.
- c. has to be used with binary code.
- d. have to be applied when determining if a result is true or false depending on a condition.
- e. have to be implemented to perform arithmetic operations on operands.



Let's Practice!

Operators in JavaScript

Instructions: Circle the correct option to complete the sentences.

JavaScript (sets up / works out / picks out) perfectly when operators are used properly. Therefore, it's important to (go ahead / go over / go on) the various types of operators and their functions in order to (clear up / set up / clear through) any doubts. Let's begin by saying that operators are used to (set up / works out / check out) values and perform arithmetic and logic operations, among other processes. They do so by (sorting out / looking forward to / choosing up) operands which refer to the values that are used in a certain process and by (choosing out / seeing into / picking out) the appropriate operator that is going to determine how those operands or values interact. Now, we need to (revised / check out / go on) what the operators and their functions are.





Extension Activity

JavaScript Operators

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for information about JavaScript operators.
- Create some flashcards where you include the type of operators and some examples.
- Choose one of the operators and prepare a presentation.
- Share your work with the class.

Self-Assessment

Yes

No

- I understand what operators in JavaScript are.
- I list the types of operators.
- I describe the characteristics of operators in JavaScripts.
- I share my work with the class.

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Target Words

- **go deeper** (expression): to explore something with more detail or to get to the bottom of a situation
e.g. Before we go deeper into the operators topic...
- **beforehand** (expression): with anticipation, before a given event
e.g. Before we go deeper into the operators topic, there are some things that should be cleared out beforehand.
- **comprise** (verb): to compose, cover or include
e.g. Assignment operators that comprise the 'simple assignment', 'add and assignment', 'subtract and assignment', and others, assign a value to the left operand taking into account the value of the right operand.
- **output** (noun): the result or final product
e.g. In case you can't get the expected output, you can rely on online tutorials or an experienced person to get you out of trouble.
- **spelling** (noun): synonym of orthography, how something is written
e.g. Make sure your spelling is correct to avoid typos and misplaced symbols.
- **work something over** (verb): to do something again
e.g. This will save you a lot of time instead of having to work something over.



Learning Points

Phrasal Verbs

Phrasal verbs are verbs that contain prepositions to create a different meaning.

Phrasal verbs are commonly used in English language. Whether it is in formal or informal conversations, they make up an important part of language usage.

Most common **phrasal verbs** are verbs that can only go with one preposition but there are some with two or three prepositions. They can also be separable or inseparable, meaning that the object can go in between the verb and the preposition or after the preposition.

Verbs working with different prepositions may have a different meaning from the verb when used with no preposition and sometimes they tend to be easily confused.

Some phrasal verbs are:

- Pick out: to choose from a wide variety
- Count on: to have the support of someone or something
- Work something over: to do something again
- Rely on: to trust something or someone

Examples

- *There are different types of operators people can pick out to perform a given action in JavaScript.*
- *It is always a good idea to count on a reference chart if you're new at JavaScript.*
- *...this will save you a lot of time instead of having to work something over.*

Bitwise Operators in JavaScript

Bitwise operators are operations that work in binary code which is a system that uses a combination of zeros and ones.

The bitwise operators allow a different organization of operands. The bitwise AND, OR, XOR, and NOT work as an operational logic between operands. Those operators run the operations in each bit or pairs of bits, or alike.

On the other hand, the shift bitwise operators reorganize the operands to interact in different ways following bit-by-bit operations.

The bitwise operators are:

- Bitwise AND
- Bitwise OR
- Bitwise XOR
- Bitwise Not
- Left shift
- Right shift
- Right shift with zero

